

D.K.T.E. Society's Textile and Engineering Institute, Ichalkaranji
(An Autonomous Institute, Affiliated to Shivaji University, Kolhapur)

Accredited with 'A+' Grade by NAAC



Department of Computer Science and Engineering
(Artificial Intelligence and Machine Learning) 2026-2027.

THE COMMUNITY ENGAGEMENT PROJECT REPORT ON

“SMART PARKING MANAGEMENT SYSTEM”

Under the guidance of: Prof. Rekha Kamble

Developed By:

1. 25UME044 : Ganesh Ujesh Raut
2. 25UAM065 : Abhishek Ajit Shirguppe
3. 25UAM069 : Sanket Subhash Patil
4. 25UAM043 : Atharv Chandrashekhar Kagale

CERTIFICATE

This is to certify that,

1. 25UME044 : Ganesh Ujesh Raut
2. 25UAM065 : Abhishek Ajit Shirguppe
3. 25UAM069 : Sanket Subhash Patil
4. 25UAM043 : Atharv Chandrashekhar Kagale

Have successfully completed the project work, entitled

“SMART PARKING MANAGEMENT SYSTEM”

In partial fulfilment for the award of degree of Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence and Machine Learning). This is record of their work carried out during academic year 2026- 27.

Prof. Rekha Kamble
[Project Guide]

Prof. Dr. S.K.Shirgave
[Head of Department]

Date:29/09/2026

Place: Ichalkaranji

DECLARATION

We the undersigned students of S.Y.C.S.E.(AIML) declare that the Community Engagement Project work report entitled “**SMART PARKING MANAGEMENT SYSTEM**” written and submitted under the guidance of Prof. Rekha Kamble is our original work. The empirical findings in this report are based on the data collected by us. The matter assimilated in this report is not reproduction from any readymade report.

Name

PRN

- | | |
|---------------------------------|----------|
| 1. Ganesh Ujesh Raut | 25UME044 |
| 2. Abhishek Ajit Shirguppe | 25UAM065 |
| 3. Sanket Subhash Patil | 25UAM069 |
| 4. Atharv Chandrashekhar Kagale | 25UAM043 |

Date: 29/09/2026

Place: Ichalkaranji

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INTRODUCTION

Parking is an important issue in crowded public places, commercial areas, hospitals, educational institutions, railway stations, shopping centres, and religious places. With the increasing number of vehicles, finding an available parking space has become difficult and time-consuming. Poor parking management can result in traffic congestion, unnecessary vehicle movement, fuel consumption, and inconvenience to visitors.

The **Smart Parking Management System** is designed to provide an efficient and organized solution to common parking problems. The system helps users identify available parking spaces and reduces the time spent searching for a vacant slot.

The main objective of the Smart Parking System is to improve the utilization and management of available parking spaces. The system can use sensors to detect whether a parking slot is occupied or vacant. The information collected by the sensors can be processed using a microcontroller and displayed to users through an LCD, LED display, or digital application.

The system can also use **IoT technology** to transmit parking information over a network. This allows parking information to be monitored in real time and can help administrators understand the current status of the parking area.

The Smart Parking System can be implemented using components such as **IR sensors or ultrasonic sensors, ESP32/Arduino microcontrollers, LCD or LED displays, Wi-Fi connectivity, and a database or cloud platform.**

The system can be particularly useful in crowded places where visitors face difficulty in finding parking spaces. By providing information about vacant slots, it can reduce unnecessary vehicle movement and improve the overall parking experience.

In conclusion, the Smart Parking System provides a practical and technology-based approach to parking management. It can help save time, improve parking-space utilization, reduce congestion, and support better management of parking facilities.

PROBLEM STATEMENT

Finding an available parking space in crowded public places is often difficult and time-consuming. Drivers may have to move around the parking area several times before finding a vacant parking slot. This can result in unnecessary fuel consumption, traffic congestion, and inconvenience.

Traditional parking systems often depend on manual supervision and do not provide real-time information about available parking spaces. Parking staff may also find it difficult to monitor a large number of vehicles and parking slots continuously.

At busy locations such as **Mahalaxmi Temple, Kolhapur**, the number of visitors and vehicles can increase significantly during weekends, festivals, and special occasions. This can create parking difficulties and traffic-management problems.

The **Smart Parking Management System** aims to address these issues by monitoring parking slots and providing information about available and occupied spaces. Sensors can detect vehicles in individual parking slots and update the parking status.

The system can help drivers locate vacant spaces more efficiently and can assist parking administrators in monitoring parking occupancy.

PROBLEM DESCRIPTION

In modern cities and public places, the increasing number of vehicles has created significant parking challenges. Drivers often spend considerable time searching for vacant parking spaces, especially in crowded locations.

In many traditional parking areas, users do not have information about parking availability before entering the parking area. As a result, vehicles move unnecessarily inside the parking area while searching for vacant spaces.

Manual parking management also requires continuous human involvement. Parking staff need to guide vehicles, observe parking spaces, and maintain information about occupied and vacant slots. This becomes difficult when the number of vehicles increases.

The major problems associated with traditional parking management include:

- Difficulty in finding vacant parking spaces.
- Unnecessary movement of vehicles.
- Increased fuel consumption.
- Traffic congestion near parking areas.
- Wastage of users' time.
- Difficulty in monitoring parking occupancy.
- Inefficient utilization of available parking spaces.
- Increased workload for parking-management staff.

The Smart Parking System addresses these problems by providing an automated method for detecting parking-slot occupancy.

Sensors can be installed at individual parking spaces to detect the presence of vehicles. The sensor information can be sent to a microcontroller, which processes the data and determines whether the parking space is available or occupied.

The current parking status can then be displayed using an electronic display or transmitted through IoT connectivity to a mobile or web application.

This system can provide useful information to both drivers and parking administrators. It can make parking management more organized and reduce the time required to search for available parking spaces.

FACILITY REQUIREMENTS RELATED TO THE PROBLEM

1. **User-Friendly Interface** :The system should provide a simple and easy-to-understand interface for users and parking administrators.
2. **Parking Slot Detection** : Sensors should automatically detect whether individual parking slots are occupied or vacant.
3. **Real-Time Parking Status** : The system should provide updated information about available and occupied parking spaces.
4. **Sensor System** : IR sensors or ultrasonic sensors can be used to detect vehicles in individual parking slots.
5. **Microcontroller** : A microcontroller such as ESP32 or Arduino can receive sensor data and process the parking information.
6. **Display System** : An LCD, LED display, or digital screen can show the number of available parking spaces.
7. **IoT Connectivity** : Wireless connectivity can be used to transmit parking information to an application or cloud platform.
8. **Database Management** :Parking information can be stored in a database for monitoring and future analysis.
9. **Admin Monitoring** : An administrator should be able to monitor the parking status and manage the parking system.
10. **Power Supply** :A reliable power supply is required for sensors, microcontrollers, displays, and communication modules.

FIELD VISIT PHOTOS



Interaction with the concerned person regarding parking management.



TECHNICAL SOLUTIONS ON PROBLEM

Several technical solutions can be implemented in the Smart Parking System to improve parking management, provide real-time information, and reduce the difficulties faced by drivers.

1. Parking Slot Detection

Description:

Parking-slot detection is the main function of the Smart Parking System. Sensors can be installed at individual parking slots to detect whether a vehicle is present.

Implementation Strategies:

- Install IR or ultrasonic sensors at each parking slot.
- Connect sensors to a microcontroller.
- Detect the presence or absence of vehicles.
- Assign a unique number to every parking slot.
- Update the slot status automatically.

Benefits:

- Automatically detects parking occupancy.
 - Provides accurate slot information.
 - Helps users locate available spaces.
-

2. Microcontroller-Based Control System

Description:

A microcontroller acts as the central control unit of the system. It receives information from parking sensors and processes the data.

Implementation Strategies:

- Use ESP32 or Arduino.
- Process sensor readings using programmed logic.
- Calculate the number of available parking spaces.
- Send information to the display or IoT platform.

Benefits:

- Provides automated parking management.
 - Processes information quickly.
 - Reduces manual work.
 - Supports multiple parking slots.
-

3. Real-Time Parking Display

Description:

A display system can provide users with immediate information about available parking spaces.

Implementation Strategies:

- Install an LCD or LED display at the parking entrance.
- Display the number of vacant slots.
- Display individual slot status when required.

Benefits:

- Helps drivers find parking quickly.
 - Reduces unnecessary vehicle movement.
 - Saves time and fuel.
 - Improves parking organization.
-

4. IoT Connectivity

Description:

IoT connectivity allows the parking system to transmit information over a network. This enables real-time monitoring of parking spaces.

Implementation Strategies:

- Use an ESP32 with Wi-Fi connectivity.
- Connect the system to an IoT platform.
- Send sensor data through the internet.
- Update parking information in real time.

Benefits:

- Enables remote monitoring.
 - Provides real-time parking information.
 - Supports mobile and web applications.
 - Allows parking data to be stored and analyzed.
-

5. Mobile Application

Description:

A mobile application can allow users to view parking availability from their smartphones.

Implementation Strategies:

- Develop an Android or cross-platform application.
- Display vacant and occupied parking slots.
- Show the location of the parking facility.
- Update parking status in real time.

Benefits:

- Provides convenient access to parking information.

- Reduces searching time.
 - Improves user experience.
 - Helps users plan parking before entering the area.
-

6. Parking Slot Identification

Description:

Every parking space can be assigned a unique identification number so that users can easily locate available spaces.

Implementation Strategies:

- Assign numbers such as P1, P2, P3, etc.
- Connect each slot with a corresponding sensor.
- Display the slot number and current status.
- Show available slots through the application.

Benefits:

- Makes parking spaces easy to identify.
 - Improves parking organization.
 - Helps users locate vacant slots.
 - Simplifies parking management.
-

7. Database Management

Description:

A database can store information about parking slots and their occupancy status.

Implementation Strategies:

- Use Firebase, MySQL, PostgreSQL, or another suitable database.
- Store parking-slot information.
- Record changes in parking status.
- Store timestamps for parking activity.

Benefits:

- Provides organized data storage.
 - Supports historical analysis.
 - Helps administrators monitor parking usage.
 - Provides data for future improvements.
-

8. Admin Monitoring System

Description:

An administrator dashboard can allow authorized personnel to monitor the complete parking area.

Implementation Strategies:

- Display the status of all parking slots.

- Show the number of available and occupied spaces.
- Monitor sensor status.
- Generate parking reports.

Benefits:

- Simplifies parking supervision.
 - Provides centralized monitoring.
 - Helps identify parking problems.
 - Improves administrative control.
-

9. Entry and Exit Management

Description:

The system can be extended to monitor vehicles entering and leaving the parking area.

Implementation Strategies:

- Install sensors at entry and exit points.
- Detect vehicle movement.
- Automatically update parking availability.
- Optionally integrate RFID or QR-code technology.

Benefits:

- Improves vehicle-flow management.
 - Automatically updates parking information.
 - Reduces manual monitoring.
 - Supports organized parking operations.
-

10. Parking Data Analysis

Description:

The collected parking data can be analyzed to understand parking requirements and usage patterns.

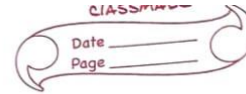
Implementation Strategies:

- Collect parking occupancy data.
- Store data with date and time.
- Identify peak parking periods.
- Generate parking-usage reports.
- Use data analysis for future planning.

Benefits:

- Helps understand parking patterns.
 - Supports better parking management.
 - Helps administrators plan parking facilities.
 - Provides a foundation for future intelligent parking solutions.
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AUTHORITY LETTER



Date : 27 /09/2026

To,
The parking area in charge,
Mahalaxmi temple,
Kolhapur, Maharashtra - 416012.

Subject: Visit to mahalaxmi temple parking
area for smart parking management system

Respected Sir,

We are student of DKTE's textile and Engineering
Institute, Ichalkaranji, visiting the mahalaxmi
temple parking area under the CEP for our
project "Smart Parking Management System."

We visited the parking area to understand the
parking facilities, vehicle entry and exist process,
parking space availability and problem faced
by visitors. The information collected will help
us understand the practical requirements of
developing a smart parking system.

Thank you for your cooperation & support.

Parking In-charge
Mahalaxmi Temple, Kolhapur.

CONCLUSION

The **Smart Parking Management System** is a technology-based solution designed to address common parking problems in crowded public places. The system can use sensors, microcontrollers, displays, IoT connectivity, and software applications to monitor parking spaces and provide real-time parking information.

The field visit to **Mahalaxmi Temple, Kolhapur**, helped us understand practical parking problems such as difficulty in finding vacant parking spaces, unnecessary vehicle movement, parking congestion, and the need for better parking management.

The proposed system can detect whether individual parking spaces are occupied or vacant and provide this information to users and parking administrators. The use of IoT technology can enable real-time transmission and monitoring of parking information.

The system can also be extended with mobile applications, databases, administrator dashboards, entry and exit monitoring, and parking-data analysis. These features can improve the overall efficiency and organization of parking facilities.

Overall, the Smart Parking System provides a practical approach to improving parking management. It can help reduce the time spent searching for parking, minimize unnecessary vehicle movement, improve utilization of available parking spaces, and provide a better parking experience for visitors.

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Research Papers / Journals:

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3. Lin, T., Rivano, H., & Le Mouël, F. (2017). *A Survey of Smart Parking Solutions*. IEEE Transactions on Intelligent Transportation Systems.

Books:

1. Bahga, A., & Madiseti, V. (2014). *Internet of Things: A Hands-On Approach*. Universities Press.
2. Raj, P., & Raman, A. C. (2017). *The Internet of Things: Enabling Technologies, Platforms, and Use Cases*. CRC Press.
3. Monk, S. (2016). *Programming Arduino: Getting Started with Sketches*. McGraw-Hill Education.

Technical Resources :

1. Arduino Documentation – Technical documentation and resources for Arduino-based systems.
2. Espressif Systems – ESP32 technical documentation and development resources.
3. Firebase Documentation – Cloud database and application development resources.